A Structural Scalar-Field Testbed: A Verification-First FEM Protocol for Medium-Based Compatibility Tests

A reproducible protocol for testing medium-based compatibility, collapse, topological spin signatures, and geometry observables across Firedrake and FEniCSx (v0.1.12 cross-backend draft)

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Abstract

"The Standard Model answers what we observe with extraordinary precision. SFT asks a different question: under what structural conditions do the observed stable configurations exist at all? This is a shift from phenomenological description to constraints on existence and stability. It is not presented as competition, but as complementarity."

Structural Field Theory (SFT) is a monistic structural-medium framework in which observed phenomena are modeled as stable configurations of a single structural field variable S. Effective phase, topological, electromagnetic, and gauge-like descriptions are extracted only under declared domain conditions, rather than introduced as freely tunable independent degrees of freedom. This *v0.1.12* draft keeps the release verification-first: it provides artifacts, schemas, hash manifests, and pre-registered pass/fail gates so independent reviewers can evaluate claims without trusting the authors. The update records a FEM bridge chain: a Firedrake/UFL reference line through v0.1.11 and an independent FEniCSx/DOLFINx cross-backend layer in v0.1.12, covering compatibility, native collapse, spin FR, double-slit readout, and Mercury/Venus perihelion gates. The v0.1.11 addendum further adds an exploratory S-rigidity compliance-window scan, mapping a bounded medium regime between rupture and persistent scalar residue under fixed physical time and locked gates. These bridge runs are not presented as global validation of SFT, nor as derivations of the Standard Model or General Relativity. They demonstrate that selected SFT-facing audit targets can be executed in an external PDE/FEM environment and reported through machine-checkable PASS/FAIL artifacts. Physical confirmation of SFT remains conditional on third-party real-solver outputs at multiple resolutions under the published gates.

Keywords:

Verification protocol; reproducibility; external audit; artifact contract; schema validation; pre-registered thresholds; computational physics; emergent phenomena; FEniCSx/DOLFINx; Firedrake; cross-backend.

1. Introduction

Many physical quantities currently enter our best models as inputs: coupling constants, particle properties, and medium-like parameters that are measured rather than derived. SFT explores an alternative stance: treat the “vacuum” as a structural medium and treat what we call matter as stable, medium-compatible configurations of a single underlying field.

This paper is deliberately verification-first: we focus on what an external reviewer can run, check, and falsify [1,2].

SFT is not presented as a replacement for the Standard Model, but as a complementary structural-medium hypothesis aimed at explaining why the observed stable configurations exist. The Standard Model remains the reference phenomenology; SFT targets a causal/mechanistic layer beneath it. Nothing in this release disputes the Standard Model’s empirical success; the intent is a reinterpretation at the level of existence/conditions, evaluated only through pre-registered gates and external runs.

• Deliverable: a protocol and tooling that minimizes trust in authors (schemas, manifests, deterministic seeds, pass/fail gates).

• Scope: EM + gravity/PPN bridge + an emergent spin-½ route; not a full Standard Model replacement [8].

• Status: verification-ready with selected Firedrake/UFL and FEniCSx/DOLFINx bridge runs; not yet globally confirmed end-to-end without independent third-party solver outputs.

Physics-Facing Audit Targets and Current Status

This release is not a narrative-only protocol: it includes representative physics-facing audit targets distributed as runnable bundles, each tied to explicit PASS/FAIL gates, SHA-256 manifests, and, where applicable, schema-validated artifacts. A compact status table is kept in the main text; the longer bundle inventory is provided in Appendix A.

To avoid overclaiming, every bundle is labeled by evidential role: CI checks verify tooling; REAL-ready modules define contracts for external solver outputs; FEM bridge runs are exploratory producer-side outputs generated in external PDE/FEM environments, currently including Firedrake/UFL and FEniCSx/DOLFINx; and REAL-verified claims remain reserved for independent third-party runs under locked gates.

Table 1. Compact evidential status of representative audit targets.

Audit layer	Representative bundle / artifact	Current status
CI audit pipeline	Double-slit synthetic gate; EOC CI; AtomX; External Review Pack; alpha-like stress test	CI (C): tooling, schemas, manifests, gates
REAL-ready contracts	Alpha-out runner; Venus/perihelion verifiers; spin Appendix S	REAL-ready (P-input): can ingest third-party outputs
FEM bridge	SFT_FIREDRAKE_CORE_REPRO_v0_1_11 + SFT_FEM_CROSSBACKEND_REPRO_v0_1_12	Exploratory P-bridge with Firedrake/UFL reference outputs and FEniCSx/DOLFINx cross-backend checks
Headline FEM results	Spin FR/topological memory; radial H/T Coulomb; Venus/Mercury perihelion; double-slit Born/readout; alpha-thread proxy; S-compliance landscape; FEniCSx compatibility/collapse/spin/double-slit/perihelion checks.	Exploratory PASS/MIXED under declared package gates; not global validation
FEniCSx cross-backend bridge	SFT_FEM_CROSSBACKEND_REPRO_v0_1_12.zip	Independent DOLFINx/FEniCSx implementation of selected compatibility, collapse, spin, double-slit, and perihelion gates.
Open roadmap branches	Alpha-Out bridge; multi-resolution third-party Firedrake/FEniCSx replication	Alpha-Out INCOMPLETE; REAL cross-backend replication pending

Interpretation.

- CI (C) means CPU-only validation of the audit pipeline (artifacts, gates, schemas, manifests). It is not end-to-end physical confirmation by itself.

- REAL-ready (P-input) means the verifier and artifact contract are in place and can ingest third-party REAL solver outputs.

- REAL-verified (P) would require an independent REAL solver run, published with artifacts + hashes, that passes the declared gates.

Note on α-out.

The packs `SFT_alpha_out_MINI_bundle_v0.3-FINAL_FIXED` and `SFT_alpha_out_MINI_runner_ready_v3_with_schema_checks_FIXED` implement the same α-out verification logic at different packaging levels; the latter adds explicit schema-checked runner readiness.

2. Minimal framework (what is being tested)

SFT assumes a single structural field variable S (dimensionless in the discrete model), interpreted as the structural/tensional degree of freedom of the medium often called "vacuum." The scalar representation is operational: effective phase, topological, spin-like, and electromagnetic descriptions are extracted from S-derived artifacts only within declared regimes and domain-of-validity checks. Matter corresponds to stable, localized or topological configurations of S; dynamics is discrete-first with a controlled continuum limit.

Interpretation (minimal): S encodes the local structural/tensional state of the medium. Stable, medium-compatible configurations of S correspond to what we describe as matter; propagating disturbances correspond to radiation modes. All statements in this paper are operationalized via exported artifacts and pass/fail gates.

2.0 Minimal physics backbone (orientation-only)

This protocol paper is intentionally verification-first; the purpose of this short backbone is to state the minimal physical model being exercised by the runners, without expanding into a full derivation (which remains in the companion technical corpus).

• State variable. S(x,t) is represented as a single structural field variable (dimensionless in the discrete model). Discrete simulations represent S on a lattice for computation; phase, topology, spin-like signatures, and gauge-like mappings are treated as extracted descriptions under declared domain conditions, not as hidden independent degrees of freedom. Physical statements are interpreted under declared domain-of-validity checks and mesh-control diagnostics.

• Minimal equation of motion (orientation). A representative continuum form is:

$\partial t^2 S - c^2\nabla^2 S + \partial V/\partial S = 0$ (often written as $\partial t^2 S - c^2\nabla^2 S + \alpha V\ S + \lambda 4\ S^3 = 0$ for the minimal quartic potential).

• Energy budget. For non-collapse evolution, the local energy density is tracked as $u = ½(\partial tS)^2 + ½c^2|\nabla S|^2 + V(S)$, with a corresponding global energy integral used for PASS/FAIL budgets. When collapse-style effective rules are enabled, any energy removed from the propagating field is recorded explicitly in a "collapse ledger" artifact to preserve auditability.

• Discrete manifestation. Although S is continuous in availability, physically relevant “objects” are defined as stable families/attractors that persist under perturbations and survive mesh refinement; particle labels (electron/proton/neutron, etc.) are treated as equivalence classes identified by invariant signatures and preregistered gates, not as a unique micro-geometry.

• Structural collapse vs measurement. Structural collapse denotes an intrinsic regime change of the medium when local propagation/relaxation limits (bounded by c and the medium response) are exceeded; it may occur in free evolution. Measurement contexts (e.g., a screen in a double-slit setup) are modeled as external interventions (apparatus-defined coupling/boundary forcing) that can trigger the same structural collapse locally, with “quantum collapse” reserved for the readout/update layer used to report outcomes.

• We state falsifiable predictions: (I) high-k modified dispersion (stringent tests of Lorentz invariance), (II) sub-mm Yukawa-like deviations in gravity, (III) cosmic birefringence, and (IV) the onset of nonlinear Breit–Wheeler pair production mapped in strong-field parameters (a_0, χ_γ) rather than a single intensity threshold. Together, these proposed tests position SFT as a reproducible, falsifiable program that bridges lattice field simulations with quantum and relativistic observables. These are proposed falsifiable tests; they are not validated in this release.

2.1 Minimal dynamics (presentation-level)

We keep the main text minimal; detailed model choices live in the accompanying corpus.

Continuum reference (for orientation):

```
L0 = 1/2 (∂μ S)(∂^μ S) − V(S)
```

Discrete runs must demonstrate controlled systematics (energy/continuity budgets, dispersion fits, convergence checks) [6,7].

2.2 Operational notion of existence (natural vs maintained)

A configuration “exists” in SFT if it is stably compatible with its medium, in a way that can be certified over a non-empty region of medium parameters under declared gates.

• Natural existence: stability holds in the ambient medium (no declared upkeep).

• Maintained existence: stability holds only when minimal declared controls (BCs, confinement, fields, filtering, etc.) are applied; ablation removes PASS.

3. Operational dictionary and audit artifacts

3.1 Compatibility operator and "existence region"

For a target configuration S and medium levers M, SFT defines a compatibility operator $\mathcal{C}$ [S;M] collecting constraints such as continuity, energy ledgers, dispersion control, EM consistency constraints (as implemented), and (when applicable) PPN mapping checks. Existence is certified by scanning or optimizing M to minimize $\|\mathcal{C}\|$ and by exporting an existence region $\mathcal{R}$ where gates PASS. Constraints are implemented as explicit verifiers over exported artifacts, not as hidden fit degrees of freedom.

3.2 Electromagnetism: derived potential and domain of validity (brief)

In this release, electromagnetism is treated as a derived description over the structural field rather than as an additional independent set of degrees of freedom. Specifically, the effective phase/angle variable θ is defined only under explicit coarse-graining assumptions (e.g., narrowband / slowly-varying regimes), and the electromagnetic potential Aμ is introduced as a functional of exported S-artifacts via a fixed Green/operator construction. Aμ is therefore not an independently tunable DOF, although it may be non-local at the potential level.

• Domain notes:

• θ-definition: valid in regimes where phase extraction is stable under declared windows (narrowband / coarse-grain). Outside those regimes, θ is treated as undefined rather than forced.

• Near $R \approx 0$ (core/defect center): EM mapping may require excision/regularization; the excision rule and window are treated as declared inputs and must be reported in SCAN metadata.

• No extra DOF: any apparent freedom in Aμ is fixed by the chosen Green/operator and boundary conditions; changes must be treated as model variants and pre-registered.

3.3 Artifact contract (SCAN → REGION → REPORT)

Any externally verifiable claim must provide a minimal artifact chain:

• SCAN: raw sweep results (CSV/JSON) with provenance (grid, BCs, solver version/hash, seed).

• REGION: a mask or list of PASS points defining an existence region $\mathcal{R}$ with at least one PASS voxel.

• REPORT: compact JSON summarizing verdict (natural/maintained/fail), thresholds, maintenance cost ΔM, robustness τ (with CI), and hashes.

JSON policy: sanitize NaN/Inf to null; enforce schema_version='1.0.0'; validate against published schemas; hash everything (SHA-256) with a manifest including a self-hash.

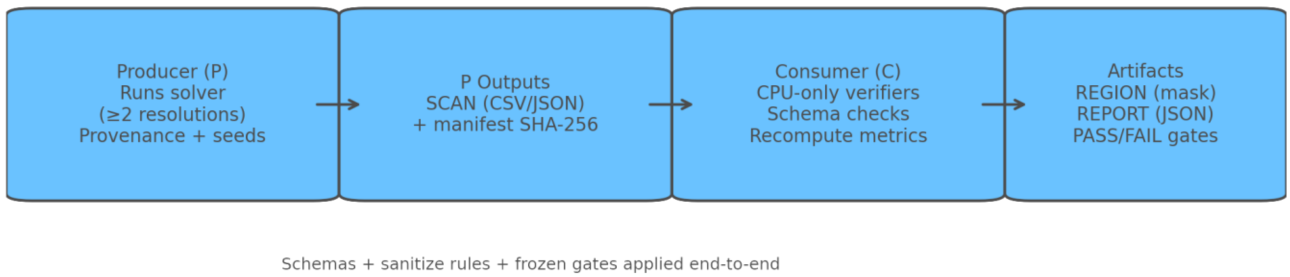

Figure 1. Verification pipeline: Producer (P) generates SCAN artifacts; Consumer (C) re-computes metrics, validates schemas/hashes, and emits REGION + REPORT under frozen gates. All PASS/FAIL results must be reproducible from SCAN alone (single source of truth).

3.4 Calibration (C) vs prediction (P) discipline

To limit degrees of freedom, SFT uses a single-pass calibration pipeline. Calibration outputs (q*, ħ*, ε*, μ*, etc.) are frozen and reused across demonstrations without per-observable retuning. Any "α-out" estimation is a separate, pre-registered exercise and does not affect the validity of α-in runs.

• α-in: α is treated strictly as an input used during EM calibration.

• α-out: optional pipeline estimating α from SFT-native observables without injecting α during calibration.

• Doc 10.1 (Alpha-Elasticity CI Suite) provides a CI/synthetic, verification-first protocol for alpha-like estimators (anti-leakage + preregistered gates) and should not be conflated with α-out.

• For limitations, preregistered acceptance bands, and the reviewer checklist (including External Break Test #01 as an anti-post-hoc audit), see Doc 8 — Limitations & Validation Plan.

4. Verification protocol and gates

This section summarizes default gates used in the distributed runner packs. Gates may be tightened, but should never be loosened post hoc for a claim. All gates must be declared in the REPORT and repeated in the REGION/SCAN metadata. For any REAL run, all thresholds must be locked (pre-registered) before solver execution and referenced by gate_id in the REPORT.

See Table 2. Detailed module gates and current evidential status

Module / claim	Status	Key P output	C verifier	Metric	Default gate	Notes (CI vs REAL)
Energy/continuity QA	REAL-ready	ENERGY_REPORT.json	verify_energy.py	$\Delta E/E$	$\Delta E/E \le 1e{-3}$	[REAL] Also report ledger/continuity residuals + convergence.
Dispersion (small-k)	REAL-ready	dispersion_smallk.csv	fit_dispersion.py	c, R^2_smallk	$R^2 \ge 0.98$; $\lvert\Delta c\rvert/c \le 1\%$	[REAL] Declare k-window; run ≥2 grids.
EOC / convergence (CI)	CI	EOC_synthetic.csv	test_eoc_ci.py	p, R^2	$1.70 \le p \le 2.05$; $R^2 \ge 0.995$	[CI] Pipeline health-check only (synthetic).
AtomX existence	CI	synthetic_atom_scan.csv	verify_atomX.py	split, compat, R^2	split ≤ 0.19 meV; compat ≤ 0.02; $R^2 \ge 0.98$	[CI] Demonstration thresholds; use ablation for maintained vs natural.
Hydrogen 2s–2p	CI	COMPATIBILITY_SCAN_H.*	verify_hydrogen.py	split, compat, R^2	split ≤ 0.10 meV; compat ≤ 0.02; $R^2 \ge 0.98$	[CI] Demonstration; REAL validation requires solver outputs + convergence.
Double slit	CI	visibility.csv, shift.csv	verify_double_slit.py	phase R^2, V monotone	$R^2 \ge 0.98$; sat ≤ 0.10; monotone↓	[CI] Synthetic by default; REAL requires solver-generated CSVs.
Venus perihelion (overlay)	REAL-ready	varpi_series.csv	verify_venus_real.py	slope, R^2	$\lvert\text{slope}-8.624984\rvert \le 0.20$; $R^2 \ge 0.999$	[REAL] Fit interval predeclared; report sensitivity.
α-out (optional)	REAL-ready	DISPERSION_LINEAR.json, HBAR_FROM_ROTOR.json, Q_EPS_SOLUTION.json	compose_alpha_out.py	tier error	Tier-1 ≤ 1%; Tier-2 ≤ 1e−4; Tier-3 ≤ 1e−6	[CI+REAL] CI validates plumbing; tiers require REAL components without α injection.

PPN bridge (optional)	REAL-ready	ppn_summary.json	verify_ppn.py		|β−1| ≤ 1e−2; |γ−1| ≤ 1e−2 (placeholder)	[REAL] Thresholds must be pre-registered (locked in a gates file/commit) before any REAL run; REPORT must reference the gate_id.
Firedrake spin bend FR	Exploratory P-bridge	spin_bend_scan.csv + SFT_SPIN_BEND_SIGNATURE_SCAN_REPORT.json	included report verifier	FR 2π phase + director closure	27/27 PASS; max distance to π/π ≈ 3.54e−15; max closure ≈ 3.71e−17	[Exploratory] Robust topological signature; not a full electron derivation.
Mercury perihelion Firedrake 1PN	Exploratory P-bridge	varpi_series_mercury.csv + SFT_SOLAR_PERIHELION_FIREDRAKE_REPORT.json	included geometry gate	slope, R^2	|slope−42.980204| ≤ 0.20; $R^2 \geq 0.995$; observed slope ≈ 43.057807, $R^2 \approx 1.0$	[Exploratory] Effective solar-system geometry gate; not a derivation of GR.
Venus perihelion Firedrake 1PN	Exploratory P-bridge	varpi_series_venus.csv + SFT_VENUS_FIREDRAKE_GEODESIC_REPORT.json	included geometry gate	slope, R^2	|slope−8.624984| ≤ 0.20; $R^2 \geq 0.995$; observed slope ≈ 8.684022, $R^2 \approx 1.0$	[Exploratory] Same bridge logic as Mercury.
Firedrake double-slit	Exploratory mixed	visibility.csv, shift.csv	included report verifier	phase R^2, visibility saturation	phase PASS; visibility saturation FAIL under sat ≤ 0.10	[Exploratory] Useful negative/hold result; not a headline validation result.
Firedrake Alpha-Out bridge	Incomplete by design	SFT_ALPHA_OUT_BRIDGE_REPORT.json	compose bridge	available c + missing components	status INCOMPLETE; missing hbar*, q*, eps*	[Roadmap] Prevents alpha-out overclaim until real components exist.

Note: An additional CI-only protocol for an **alpha-like estimator** (anti-leakage, preregistered decision rule hashing, null/adversarial controls) is provided in **Doc 10.1**. This is a verification methodology appendix and is not a physical α validation.

Important: CI bundles certify the verification pipeline only; physical claims require REAL solver outputs under the published gates.

Synthetic CI modules certify tooling integrity (schemas, manifests, regression gates) in CPU-only environments. Scientific claims require P outputs produced by a real solver on at least two resolutions under the declared gates.

5. Alpha as an effective structural thread metric

We adopt a monistic structural-medium view: all observables arise as stable configurations of a single field S. In this context, dimensionless couplings may be interpreted as normalized material properties of the underlying medium rather than arbitrary numbers.

We define a measurable stiffness proxy via the energetic response to controlled perturbations. For a baseline state S0, impose $\delta S(x)=\varepsilon \cos(k\cdot x)$ and compute the energy increment $\Delta E(\varepsilon,k)$. The effective spectral stiffness is $K_{eff}(k) = (1/V) * \partial^2 \Delta E/\partial\varepsilon^2 |_{(\varepsilon=0)}$. This definition is operational and directly auditable from solver outputs.

In SFT the fine-structure constant is expressed as alpha = q_star^2/(4*pi*epsilon_star*hbar_star*c), highlighting alpha as a dimensionless ratio between an emergent electrostatic scale (q_star^2/4*pi*epsilon_star) and the medium's action-propagation scale (hbar_star*c). In the current wording, alpha is not treated as a primitive cause or as the compliance of the medium itself. It is treated as an externally measured effective thread/fit metric between an electron-like localized configuration and a photon-like radiative configuration, constrained by the medium response. An extended note (definitions, dimensional sanity, and external-verification contract) is provided in Appendix B.

$$K_{eff}(k) = (1/V) * \partial^2\Delta E/\partial\varepsilon^2 \,|_{(\varepsilon=0)}$$

$$\alpha = q^{*2} / (4\pi\, \varepsilon^* \,\hbar^*\, c)$$

6. External verification contract

6.1 Producer (P): what third parties must generate

To evaluate SFT claims, a third party must generate solver outputs with full provenance and at least two resolutions:

- Solver provenance: version/hash, grid(s), Δt/CFL, BCs, seeds, and any AMR thresholds.
- Integrity: MANIFEST_SHA256.txt with SHA-256 of all exported files (plus self-hash) [3].

6.2 Consumer (C): CPU-only verification

- Verify hashes and schema compliance (schema_version='1.0.0', JSON sanitize) [3-5].
- Recompute metrics and PASS masks from the SCAN artifacts (single source of truth).

• Emit REGION and REPORT artifacts with explicit thresholds and PASS/FAIL outcomes.

• External Firedrake/UFL and FEniCSx/DOLFINx bridge mini-packs are also included; see Appendix A.

7. FEM bridge runs: Firedrake/UFL reference plus FEniCSx/DOLFINx cross-backend audit targets

Parameter thread across FEM bridge modules. The bridge results reported below preserve SFT_FIREDRAKE_CORE_REPRO_v0_1_11 as the Firedrake-first reference and add SFT_FEM_CROSSBACKEND_REPRO_v0_1_12 as an independent FEniCSx/DOLFINx cross-backend layer. The package does not impose one universal physical parameter value across heterogeneous tests; instead, each module carries its own declared medium/discretization block in its JSON report (mesh, polynomial degree, boundary conditions, thresholds, and body-specific constants where applicable). The common constraint is protocol-level: no per-observable retuning after gates, schema_version='1.0.0', SHA-256 manifests, and machine-checkable PASS/FAIL gates. This is the parameter thread connecting the Firedrake and FEniCSx bridge outputs.

This section records the external-PDE/FEM bridge evidence chain for SFT-facing audit targets. Firedrake/UFL provides the first reference implementation; FEniCSx/DOLFINx now provides an independent cross-backend check on selected gates. The package is not treated as a global validation of SFT. Its narrower contribution is to show that selected medium-field diagnostics can be generated in external FEM environments and exported as auditable JSON/CSV artifacts with explicit PASS/FAIL outcomes.

7.1 Package status and integrity

The package `SFT_FIREDRAKE_CORE_REPRO_v0_1_10` contains the consolidated Firedrake bridge outputs, and `SFT_FIREDRAKE_CORE_REPRO_v0_1_11` adds the S-rigidity compliance landscape. The cross-backend package `SFT_FEM_CROSSBACKEND_REPRO_v0_1_12` preserves the Firedrake reference and adds FEniCSx/DOLFINx modules for the compatibility operator, native collapse, double-slit readout, Mercury/Venus perihelion, and spin FR gates. Package-level manifests and per-module manifests are included for audit.

Interpretation rule. A Firedrake or FEniCSx bridge PASS means that the exported artifact passed its declared gate inside that exploratory external solver environment. It does not by itself establish end-to-end SFT physical validation, nor does it remove the requirement for independent third-party runs at multiple resolutions.

7.2 Headline bridge results

Module	Artifact/report	Observed result	Gate status	Claim boundary
Spin bend FR robustness	SFT_SPIN_BEND_SIGNATURE_SCAN_REPORT.json	27/27 cases PASS; max distance to π/π ≈ 3.54e−15; max director closure ≈ 3.71e−17	PASS	Robust spin-1/2 FR bend signature; not a full electron derivation.
Radial Coulomb H/T 2s-2p	SFT_RADIAL_COULOMB_FIREDRAKE_REPORT.json	H and T 2s/2p degeneracy PASS; max split approx 1.24e-5 meV; max absolute energy error approx 1.43e-5 meV under 1.0 meV gate	PASS	Hydrogenic radial boundary/topology sanity gate; not an SFT derivation of atoms.
Mercury perihelion 1PN	SFT_SOLAR_PERIHELION_FIREDRAKE_REPORT.json	slope approx 43.057807 arcsec/century vs 42.980204 expected; R2 approx 1.0	PASS	Solar geometry gate; not a derivation of GR or SFT gravity.
Venus perihelion 1PN	SFT_VENUS_FIREDRAKE_GEODESIC_REPORT.json	slope ≈ 8.684022 arcsec/century vs 8.624984 expected; R^2 ≈ 1.0	PASS	Same geometry gate under Venus parameters.
Alpha localization capacity scan	SFT_ALPHA_LOCALIZATION_SCAN_REPORT.json	alpha_crit bracket [1.55, 1.6]; monotonic severity criteria PASS	PASS	Medium-capacity localization scan; not particle identification.
Double-slit Firedrake	SFT_DOUBLE_SLIT_FIREDRAKE_REPORT.json	phase R^2 = 1.0 PASS; visibility saturation FAIL (last-step rel. change ≈ 0.274 > 0.10)	MIXED / HOLD	Useful negative/hold result; keep separate from synthetic CI PASS.
Alpha-Out bridge	SFT_ALPHA_OUT_BRIDGE_REPORT.json	status INCOMPLETE; missing hbar*, q*, eps*	INCOMPLETE	Roadmap artifact; prevents false alpha-out claim.

Scalar recovery/residue + spin topological memory	SFT_RECOVERY_RESIDUE_2D_SCAN_REPORT.json + SFT_SPIN_TOPOLOGICAL_MEMORY_REPORT.json	Scalar branch disperses without persistent scalar residue under declared gates; spin/director branch reports topological memory PASS across the 27-case robustness scan.	PASS / CONTROL	Paired negative/positive control: scalar residue is not the source of the spin memory signal; not a particle derivation.
Double-slit Born readout	SFT_DOUBLE_SLIT_BORN_READOUT_REPORT.json	Born detector readout PASS: TV distance approx. 0.0213, max bin error approx. 0.00308, chi2/bin approx. 0.976 for 20,000 events.	PASS	Effective readout over a fixed coherent field; not a derivation of Born's rule or a collapse simulation.
FEniCSx compatibility + native collapse	SFT_FENICSX_COMPATIBILITY_OPERATOR_REPORT.json + SFT_FENICSX_NATIVE_COLLAPSE_REPORT.json	Compatibility smoke GLOBAL_PASS; native collapse gate crossed with first_collapse_step 3, collapsed_dofs_final 17, max_T00_seen 0.657675, and T00/rho_crit approx 1.315.	PASS / EXPLORATORY	Independent FEM translation of the SFT compatibility/collapse logic; not a particle derivation or physical validation.
FEniCSx double-slit phase/visibility/Born	SFT_FENICSX_DOUBLE_SLIT_REPORT.json	Grid32 smoke and grid64 minimap report phase R2=1.0, monotone visibility, Born TV approx 0.0304, max bin error approx 0.00809, chi2/bin approx 0.839.	PASS	Effective coherent-field/readout gate in FEniCSx; Born rule is applied as readout, not derived.
FEniCSx Mercury/Venus perihelion 1PN	SFT_FENICSX_SOLAR_PERIHELION_REPORT.json	Mercury slope approx 43.057807 vs 42.980204 expected; Venus slope approx 8.684022 vs 8.624984 expected; R2 near 1.0.	PASS	Cross-backend geometry sanity gate; not a derivation of GR or SFT gravity.

FEniCSx spin FR	SFT_FENICSX_SPIN_FR_REPORT.json	berry_phase_over_pi approx 1.0; distance_to_pi_over_pi approx 5.94e-15; director_l2_closure approx 1.37e-32; final spinor overlap real approx -0.9986.	PASS	Independent topological bend-signature check; not a full electron derivation.
Alpha-thread proxy gate	SFT_ALPHA_THREAD_GATE_REPORT.json / thread_metric_autotune reports	Grid64/96/128 scans report 108/108 candidate cases and GLOBAL_PASS=true; structural thread metric remains stable across grids.	PASS / PROXY	Structural fit/thread proxy only; not a derivation of the electron, photon, or physical fine-structure constant.
S-compliance landscape / S-rigidity window	SFT_S_RIGIDITY_LANDSCAPE_REPORT.json	Focused nx=128 fixed-time scan: 20 cases; 9 comfortable, 5 rupture, 6 residue/unstable, 0 unavailable; comfortable bracket kappa=0.40-0.42, amp=0.30-0.35.	PASS / EXPLORATORY	Medium-compliance map bounded by rupture and scalar residue; not an alpha result or particle claim.

7.3 Recommended interpretation

The cleanest publication claim is therefore: the verification-first contract has now been exercised by a Firedrake/UFL reference bridge and an independent FEniCSx/DOLFINx cross-backend bridge on selected modules. The strongest current bridge results are the spin bend FR robustness scan, the FEniCSx spin FR reproduction, the spin topological memory control, the radial Coulomb H/T 2s/2p sanity gate, the Venus/Mercury perihelion geometry gates in both FEM lines, the double-slit Born-readout gate, the FEniCSx double-slit phase/visibility/Born smoke/minimap, the exploratory alpha-thread proxy gate, and the S-rigidity compliance-window scan. The Firedrake double-slit phase/visibility branch should still be reported honestly as mixed under its strict visibility saturation gate; the Born readout is a separate effective detector-readout result, not a derivation of Born's rule. Alpha-thread is a structural fit proxy, not a derivation of the electron, photon, or physical fine-structure constant. The S-rigidity landscape is likewise a medium-compliance map, not an alpha result.

7.4 Implications for the roadmap

The immediate next milestones are: (I) repeat headline bridge modules at at least two resolutions with locked gates in both Firedrake and FEniCSx where feasible; (II) preserve v0.1.11 as the Firedrake reference and v0.1.12 as the cross-backend bridge; (III) generate real alpha-out components (dispersion.csv, HBAR_FROM_ROTOR.json, coulomb_profile.csv, and field_energy.json) without alpha injection; and (IV) keep all FAIL or HOLD branches visible rather than tuning thresholds post hoc.

7.5 Alpha-thread and Born-readout consolidation (v0.1.10; exploratory)

After the v0.1.9 bridge package, the post-run alpha-thread material and double-slit Born-readout material were consolidated into SFT_FIREDRAKE_CORE_REPRO_v0_1_10. The alpha-thread component preserves the stricter proxy gate R[e_proxy, gamma_proxy], where the localized side carries a director/spin FR signature and the radiative side is placed at a natural same-distortion-level amplitude before measuring a structural thread metric. The Born-readout component samples detector events from the normalized intensity of the coherent double-slit field.

The saved 64, 96, and 128 grid runs each report 108/108 candidate cases and GLOBAL_PASS=true under the declared exploratory gate. The structural thread metric ranges were 0.349711-0.619042 on grid64, 0.341954-0.605904 on grid96, and 0.336752-0.597070 on grid128. The same best and weakest parameter cases were selected across all three grids. This is a numerical stability result for the proxy gate, not a derivation of the electron, photon, or fine-structure constant.

The package SHA-256 is 4643B7D9CC2F4AE715BE42805065509ABD2EC743FAFD9A61BB460683C3F59F03. It should be cited, if at all, as an internal exploratory addendum pending independent reproduction and a clearer physical mapping.

7.6 v0.1.10 scope note

The v0.1.10 package should be read as a consolidation release rather than as a new physical claim. It joins the v0.1.9 bridge suite with the post-v0.1.9 Born-readout and alpha-thread exploratory gates. Firedrake remains an independent finite-element substrate; the SFT content resides in the compatibility operators, gates, topological/readout rules, and declared decision ledgers layered on top.

Recommended wording: SFT_FIREDRAKE_CORE_REPRO_v0_1_10 consolidates selected Firedrake/UFL bridge tests, including spin/topological memory, radial and perihelion sanity gates, double-slit Born readout, recovery/residue controls, and an exploratory alpha-thread proxy gate. It does not prove SFT, derive Born's rule, derive the electron/photon, or derive the physical fine-structure constant.

7.7 S-compliance landscape addendum (v0.1.11; exploratory)

The v0.1.11 addendum freezes a focused S-rigidity landscape result rather than opening a new alpha claim. The question is deliberately upstream: where is the comfortable rigidity/compliance band of the S medium? The scan varies kappa and impulse amplitude amp under fixed physical time T_final=3.125, cfl=0.0625, and locked gates. Cases are classified as comfortable compatibility, rupture, residue/unstable, subcritical/decoupled, or unavailable. Alpha is explicitly excluded from the decision rule.

The focused nx=128 run reports 20 total cases: 9 comfortable, 5 rupture, 6 residue/unstable, and 0 unavailable. The comfortable bracket is kappa=0.40-0.42 and amp=0.30-0.35; the best representative comfortable case is kappa=0.42, amp=0.30. The package hash is SHA-256 8F3062FED7F6A87157581C9FAC8DB19ADFE03F786DE9A5DAA291A1376EAA1E0F.

Interpretation. This is a medium-compliance map: inside the comfortable band the run avoids dynamic rupture and persistent scalar residue under the declared gates; the high-amplitude edge ruptures, while the higher-kappa edge crosses into persistent scalar residue just above the preregistered 0.10 residue threshold. The result should be cited as evidence that the Firedrake translation can expose a bounded S-compliance window, not as a derivation of alpha, a particle claim, or a global validation of SFT.

Recommended wording: SFT_FIREDRAKE_CORE_REPRO_v0_1_11 adds an exploratory S-rigidity compliance-window scan. It maps a declared Firedrake medium regime with fixed physical time and locked gates, finding a comfortable band bounded by rupture and scalar residue. It does not fit alpha, target 1/137, derive electron/photon physics, or replace independent third-party REAL verification.

7.8 FEniCSx cross-backend bridge (v0.1.12; exploratory)

This subsection integrates the v0.1.12 FEniCSx/DOLFINx bridge into the main Section 7 evidence chain, rather than treating it as a post-DOI addendum. Firedrake v0.1.11 remains the frozen Firedrake-first reference; SFT_FEM_CROSSBACKEND_REPRO_v0_1_12.zip adds an independent FEM implementation layer for selected gates.

Cross-backend package: SFT_FEM_CROSSBACKEND_REPRO_v0_1_12.zip; SHA-256 8CC5650778C0526A96253188AEA3D4BF035287B147875FE53628BE9A6DA90533. Package role:

preserve the Firedrake reference while adding FEniCSx modules, outputs, protocols, and manifests for independent audit.

FEniCSx bridge modules: compatibility_operator_2d.py; native_collapse_2d.py; double_slit_fenicsx.py; solar_perihelion_fenicsx.py; spin_fr_fenicsx.py.

Added FEniCSx results: compatibility operator GLOBAL_PASS with 11/12 compatible_maintained and empty no-drive case decoupled; native collapse gate crossed dynamically; Mercury and Venus 1PN perihelion gates PASS; spin FR reports a pi Berry/FR signature with director closure; double-slit phase, visibility, and Born-readout gates PASS in the FEniCSx smoke/minimap runs.

Interpretation. The cross-backend results show that the SFT medium-based compatibility formulation is computationally explicit across selected FEM environments for declared gates. They do not prove SFT, establish the physical existence of the medium, derive the Standard Model, derive General Relativity, or establish matter programmability.

Recommended wording: v0.1.12 is a cross-backend audit bridge. It supports the narrower claim that selected SFT-facing compatibility, collapse, topological, readout, and geometry gates can be represented reproducibly in both Firedrake/UFL and FEniCSx/DOLFINx under declared protocols.

8. Limitations and roadmap (explicit)

We adopt strict scope and labeling to avoid overclaim.

• Experimental validation is pending for several proposed tests; this release focuses on making validation short and auditable.

• Scope is currently limited to EM + gravity mapping and an emergent spin-½ route; weak/flavor/hadronic completeness remains future work.

• Discrete vs continuum tension is treated as controlled systematics: O((a k)^2) corrections are measured and reported via dispersion and EOC scans.

• α-out is optional and must be reported separately from α-in calibration; it is a falsifiable add-on via predeclared tiers.

Roadmap (verification-first):

1. Complete a Global Consistency Index (GCI) across all validated systems under a single frozen parameter set.
2. Add tests/verifiers for EM mapping uniqueness/domain (fixed operator + gauge fixing; no extra DOF; BC sensitivity).
3. Replicate the Venus/Mercury FEM perihelion gates independently across Firedrake and FEniCSx at ≥2 resolutions with locked gates and published manifests.

4. Publish a minimal GPU run recipe for third parties (config + expected output files + gates) per module.
5. Quantum collapse (readout) is treated as an effective rule in this release.

8.1 Known failure modes (reviewer checklist)

To keep the evaluation robust and non-permissive, reviewers should treat the following as common failure modes and reject runs that exhibit them unless explicitly pre-registered and justified.

• Leakage (features vs targets): any predictor derived using the target observable (directly or via rescaling) invalidates the claim. Use strict C vs P discipline and group-wise LOO where applicable.

• Post-hoc gate changes: thresholds must be declared before solver runs. Tightening is allowed only if applied retroactively to all runs; loosening is disallowed.

• Insufficient resolution / pre-asymptotic regime: claims require ≥2 resolutions and evidence of convergence (or an explicit pre-asymptotic analysis).

• Boundary-condition artifacts: show BC sensitivity (or mitigation) and document BC choices in SCAN metadata; reject results that vanish under modest BC perturbations.

• Non-reproducible PASS masks: PASS/FAIL must be reconstructible from SCAN alone (single source of truth) using the published verifiers and schemas.

9. Conclusion

We released SFT as a verification-ready protocol: a minimal operational dictionary plus externally auditable artifacts (SCAN -> REGION -> REPORT), schemas, hash manifests, and preregistered pass/fail gates. This draft now records a FEM bridge chain: a Firedrake/UFL reference package through v0.1.11 and an independent FEniCSx/DOLFINx cross-backend bridge in v0.1.12. The strongest bridge results are the robust spin bend FR scan, the FEniCSx spin FR reproduction, the spin topological memory control, the radial Coulomb H/T 2s/2p sanity gate, the Venus/Mercury perihelion geometry gates, the double-slit Born-readout and FEniCSx phase/visibility/Born gates, the exploratory alpha-thread proxy gate, and the S-rigidity compliance-window scan. The Firedrake double-slit phase/visibility branch remains deliberately reported as mixed under its strict visibility gate, alpha-thread remains a proxy metric rather than a physical derivation of alpha, and the S-compliance landscape remains a map of medium regimes rather than a particle derivation. The present status is therefore more informative than a pure CI protocol or a single-backend bridge, but still not a global validation of SFT. Physical confirmation requires independent third-party REAL solver outputs at multiple resolutions under the published gates.

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Appendix A. Notation and conventions (short)

• Signature and units; normalization of S in the discrete model.

• Definition of c from small-k dispersion; definition of correlation length if used.

• PPN mapping conventions if the gravity bridge is tested.

Hashing standard: see Ref. [3].
Schema standard: see Ref. [5].

Bundles included (representative):

• External Review Pack [CI] (ALL_CHECKSUMS): Top-level reviewer bundle: schema validation + manifest checks + demo SCAN→REGION→REPORT artifacts.

• AtomX [CI] (Designed matter demo): Step-by-step recipe + CI scan/region/report; illustrates natural vs maintained via declared controls/ablation.

- FEM bridge v0.1.10/v0.1.11/v0.1.12 [exploratory P-bridge]: Firedrake/UFL provides the reference executable mini-pack through the S-compliance landscape; SFT_FEM_CROSSBACKEND_REPRO_v0_1_12 adds an independent FEniCSx/DOLFINx bridge for selected compatibility, native-collapse, spin, double-slit, and perihelion gates. The bridge is an audit substrate, not a proof of SFT.

• Hydrogen radial [CI] (Phase-2 golden): Numerical QA bundle: 2s–2p degeneracy gate, normalization + virial checks (solver correctness sanity test).

• EOC CI synthetic [CI]: CPU-only regression test ensuring observed order-of-convergence tooling behaves correctly (synthetic smoke test).

• Optics null tests [CI]: Birefringence/dispersion null-analysis gates; demonstrates proceed/hold logic and stability checks.

• Double-slit verifier [CI]: Phase-fit and visibility saturation gates; synthetic-by-default CI harness + hooks for real solver CSVs.

• Venus perihelion verifier + real overlay [REAL-ready]: Extraction/fit/verification pipeline for perihelion slope; ready to consume third-party orbit outputs.

• Spin Appendix S runner [REAL-ready]: Fermi-rotation ($2\pi \rightarrow \pi$ phase) + rotor spectrum gates; robustness requires ≥2 grids and ≥2 seeds in REAL runs.

• α-out runner [REAL-ready] (with schema checks): Optional pipeline composing α from dispersion, rotor, and Coulomb/energy components; CI validates plumbing; REAL needed for tiers.

• Alpha-Elasticity CI Suite [CI] (Doc 10.1): Synthetic verification protocol for an alpha-like estimator (DECISION_RULE.json + decision_rule_sha256 + null/adversarial gates + manifest integrity). Demonstrates PASS/FAIL audit semantics; not a physical α claim.

Quickstart (reviewer):

1) Verify hashes → 2) Validate schemas → 3) Run verifiers on provided artifacts → 4) Swap synthetic CI inputs for real solver outputs (P) and re-run gates.

Recommended minimal external (P) deliverables for a 'REAL' verification run:

• At least two resolutions (grids) per module and, where feasible, at least two FEM backends; report convergence or GCI.

• Full provenance (solver version/hash, BCs, Δt/CFL, seed(s), platform).

• Module outputs exactly as named in Table 2, Section 7.2, and Section 7.8 (or documented equivalents), plus MANIFEST_SHA256.txt.

• No post-hoc gate changes: thresholds must be declared before running the solver.

Appendix B. Alpha note (extended)

This appendix contains the longer, reviewer-safe note supporting Section 5. It is included for completeness and for external verifiers who wish to implement α-out and stiffness diagnostics.

Purpose. This note rewrites the intuition "alpha measures the rigidity of the S medium" in a reviewer-safe way: dimensionally consistent, operationally defined, and externally verifiable without requiring the author to run GPU-scale simulations.

1. Paper-ready text (3 paragraphs)

We adopt a monistic structural-medium view: all observables arise as stable configurations of a single field S. In this context, dimensionless couplings are naturally reinterpretable as normalized material properties of the underlying medium rather than as arbitrary numbers.

We define a measurable stiffness proxy of the medium through its energetic response to controlled perturbations. For a baseline state S0, impose $\delta S(x)=\varepsilon \cos(k\cdot x)$ and compute the total energy increment $\Delta E(\varepsilon,k)$. The effective spectral stiffness is defined as $K\_eff(k) = (1/V) * \partial^2\Delta E/\partial\varepsilon^2 |\_(\varepsilon=0)$. This definition is operational and directly auditable from solver outputs.

In SFT the fine-structure constant is expressed as alpha = q_star^2/(4*pi*epsilon_star*hbar_star*c). This highlights alpha as a dimensionless ratio between an emergent electrostatic scale (q_star^2/4*pi*epsilon_star) and the structural action-propagation scale (hbar_star*c) of the medium. In the present reviewer-safe framing, alpha is not identified with a dimensional modulus or with the medium compliance itself; it is treated as an externally measured effective thread/fit metric constrained by the medium response. Its quantitative value is delegated to external verification via the alpha-out pipeline.

2. Definitions and dimensional sanity

Key point: alpha is dimensionless. A Young's modulus E has dimensions (energy/volume). So the correct statement is not "alpha equals Young's modulus", but that alpha can parameterize a normalized stiffness/compliance once a reference scale is fixed by the model's dimensional constants (a, t0, kappa, ħ*, q*, etc.).

3. Operational stiffness proxy (external-verification friendly)

Baseline: choose a background state S0 (vacuum or specified steady background).

Perturbation: $\delta S(x) = \varepsilon \cos(k\cdot x)$ with small ε and chosen wavevector k.

Measurement: for each (ε,k), compute total energy $E(\varepsilon,k)$ from the solver's energy functional.

Definition: $\Delta E(\varepsilon,k) = E(\varepsilon,k) - E(0,k)$. Then define:

$K\_eff(k) = (1/V) * (\partial^2\Delta E/\partial\varepsilon^2)$ evaluated at $\varepsilon = 0$.

Practical estimation: evaluate ΔE at a small symmetric set of ε values (e.g., $\pm\varepsilon 1$, $\pm\varepsilon 2$) and fit a quadratic in ε. Report K_eff(k) with confidence intervals and convergence checks across at least two grid resolutions.

4. Length scales without ad hoc assumptions

If the linearized dynamics around S0 admits a dispersion $\omega^2 \approx c^2 k^2 + m\_eff^2$, then a natural correlation length is:

$\xi = c / m\_eff$.

Here m_eff^2 can be estimated either from the curvature of the effective potential around S0 ($V''(S0)$) or from a small-k dispersion fit. This avoids asserting $\xi = 1/\text{alpha}$ without derivation.

5. What "alpha as a thread metric" means in SFT (reviewer-safe wording)

Because alpha = q_star^2/(4*pi*epsilon_star*hbar_star*c), alpha quantifies how strongly the emergent electrostatic scale competes with the medium's action-propagation scale. Interpreting alpha as an effective thread/fit metric is therefore a statement about an externally measured ratio of compatible structural scales, not an identification with a dimensional modulus or a primitive compliance.

Optional (one-sentence intuition): smaller alpha corresponds to a medium that is harder to polarize into EM response per unit structural action–propagation, in the chosen normalization.

6. External verification contract (minimal P outputs + C verifiers)

To validate the interpretation with third-party compute, separate:

P (producer): generates solver outputs on target grids/BCs.

C (consumer): runs CPU-only verifiers that compute K_eff(k), ξ, and alpha-out components.

Minimum producer outputs (examples):

energy_vs_eps_k.csv (or JSON) with E(ε,k) on at least two grids.

dispersion_smallk.csv with ω(k) for small k on matching grids.

coulomb_profile.csv and field_energy.json (if composing q*, ε*), using declared windows/cutoffs.

rotor_spectrum.csv plus I(S) report (if composing ħ*).

Minimum consumer verifiers (CPU-only):

fit_stiffness.py: estimates K_eff(k) + CI + convergence summary.

fit_dispersion.py: estimates c and m_eff (and thus ξ) + CI.

compose_alpha_out.py: composes alpha-out with tier thresholds.

7. Framing notes (avoid overclaim)

Use “interpret” / “parameterizes” / “is consistent with” rather than “equals” when speaking about elastic moduli. Mark toy profiles (e.g., hedgehog) as illustrations, not derivations. Reserve “validated” for cases where producer (P) outputs from the real solver are provided and pass the published gates.

Data Availability Statement

The data and verification artifacts supporting this study are available in the project repository (see "Repository" section) and in the associated release assets/manifests.

Repository structure

- FEM bridge v0.1.10/v0.1.11/v0.1.12 [exploratory P-bridge]: Firedrake/UFL provides the reference executable mini-pack through the S-compliance landscape; SFT_FEM_CROSSBACKEND_REPRO_v0_1_12 adds an independent FEniCSx/DOLFINx bridge for selected compatibility, native-collapse, spin, double-slit, and perihelion gates. The bridge is an audit substrate, not a proof of SFT.

Repository Github: https://github.com/Xaquer69/sft-theory-and-runners MIT License

Zenodo DOI: https://doi.org/10.5281/zenodo.21258531